What changed in skill forecasting—and how much?

Tactical observer B01 → frozen trained pilot A01 | Exact +1, +2 and +5 s selections

55.90 → 63.48%

+5 s TOP-1

+7.58 percentage points
17.19% fewer top-1 errors

0.274 → 0.366

+5 s MACRO-F1

Equal weight across all 33 labels
The gain extends beyond overall accuracy

THE COMPARISON

Training frequency → original GRU / transformer → improved 12-model ensemble

The prediction target and evaluation population stay fixed

Same 33-label vocabulary, same pilots, same scenarios and same missing-label masks.

ONE-SIDED INPUT

B01 observations available through time t.
No pilot ID or private selected label as input.
Labels are the exact choices at t+1, t+2, t+5.

GROUPED DATA

3,000 episodes: 300 scenarios × 10 pilots.
210 train / 45 validation / 45 test groups.
All ten pilot encounters stay in one group.

Forecast horizon	Valid validation origins	Labels with truth support
+1 s	30,802	26 of 33
+2 s	30,466	26 of 33
+5 s	29,458	26 of 33

The improved system has a larger information budget

The experiment compares complete forecasting procedures, not architecture alone.

Setting	Original GRU	Original transformer	Improved ensemble
Observation history	5 seconds	5 seconds	20 / 60 seconds
Features per token	54	54	91
Neural architecture	2-layer GRU	2-layer transformer	4 temporal MLP variants
Parameters	234,723	439,651	28,977,060 total
Seeds / fitted models	Seed 42 / one	Seed 42 / one	42, 43, 44 / twelve
Checkpoint selection	Validation NLL	Validation NLL	Training-only partitions
Selected epochs	15	20	46 / 32 / 40 / 40

Frequency reference: training-label priors only; no observation history or neural network.

Read accuracy, class balance and confidence together

Each metric answers a different question; a single ranking can hide important trade-offs.

Metric	Question it answers
Top-1 / top-3 ↑	Is the true label first / among the three highest probabilities?
Macro-F1 (33) ↑	How well are labels recovered when every catalogue entry counts equally?
Supported macro-F1 (26) ↑	The same F1 average, restricted to labels with truth support.
Negative log-likelihood ↓	How much probability is assigned to the actual label?
Multiclass Brier score ↓	How far is the entire probability vector from the correct one-hot target?
Calibration error (ECE) ↓	How close is top-label confidence to observed accuracy in fixed bins?

+1 second: all four methods on the same origins

30,802 valid origins | Higher accuracy / F1, lower NLL are better.

Model	Top-1 ↑	Top-3 ↑	F1 (33) ↑	F1 (26) ↑	NLL ↓
Training frequency	17.84%	50.03%	0.0092	0.0116	2.3921
Original GRU	54.40%	86.95%	0.2374	0.3013	1.2606
Original transformer	58.97%	88.40%	0.3010	0.3820	1.2216
Improved ensemble	65.62%	90.10%	0.4120	0.5230	1.1155

IMPROVEMENT OVER THE ORIGINAL TRANSFORMER

Top-1: +6.65 points · Top-3: +1.70 points

Macro-F1: +0.1110 · NLL: -0.1061

+2 seconds: all four methods on the same origins

30,466 valid origins | Higher accuracy / F1, lower NLL are better.

Model	Top-1 ↑	Top-3 ↑	F1 (33) ↑	F1 (26) ↑	NLL ↓
Training frequency	17.99%	50.41%	0.0092	0.0117	2.3899
Original GRU	54.01%	86.72%	0.2309	0.2931	1.2768
Original transformer	58.38%	88.22%	0.2953	0.3748	1.2361
Improved ensemble	65.17%	89.82%	0.4115	0.5223	1.1324

IMPROVEMENT OVER THE ORIGINAL TRANSFORMER

Top-1: +6.79 points · Top-3: +1.60 points

Macro-F1: +0.1162 · NLL: -0.1037

+5 seconds: all four methods on the same origins

29,458 valid origins | Higher accuracy / F1, lower NLL are better.

Model	Top-1 ↑	Top-3 ↑	F1 (33) ↑	F1 (26) ↑	NLL ↓
Training frequency	18.46%	51.48%	0.0094	0.0120	2.3799
Original GRU	51.56%	85.04%	0.2123	0.2694	1.3363
Original transformer	55.90%	86.51%	0.2739	0.3476	1.2903
Improved ensemble	63.48%	89.25%	0.3665	0.4651	1.1642

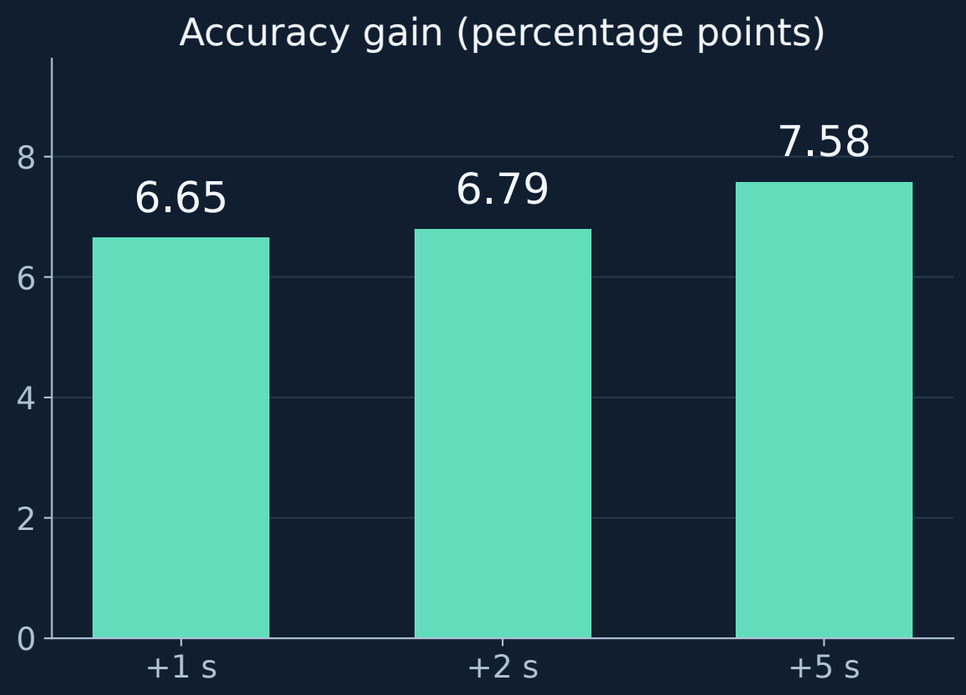
IMPROVEMENT OVER THE ORIGINAL TRANSFORMER

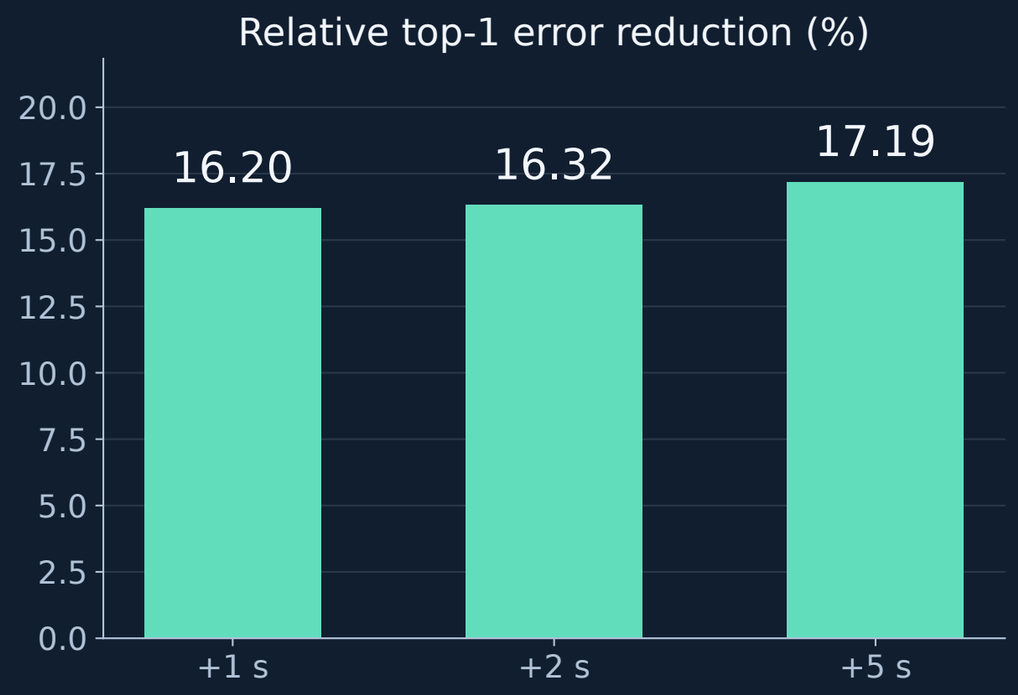
Top-1: +7.58 points · Top-3: +2.73 points

Macro-F1: +0.0926 · NLL: -0.1261

Percentage-point gain is not relative error reduction

Both improve, but their denominators differ.





+5 s: error rate falls from 44.10% to 36.52%; 7.58 ÷ 44.10 = 17.19% relative reduction.

Derived from matched top-1 accuracies | Development validation; test reserved

The ensemble fixes errors and introduces new ones

Paired outcomes on the same forecast origins reveal what the net accuracy gain hides.

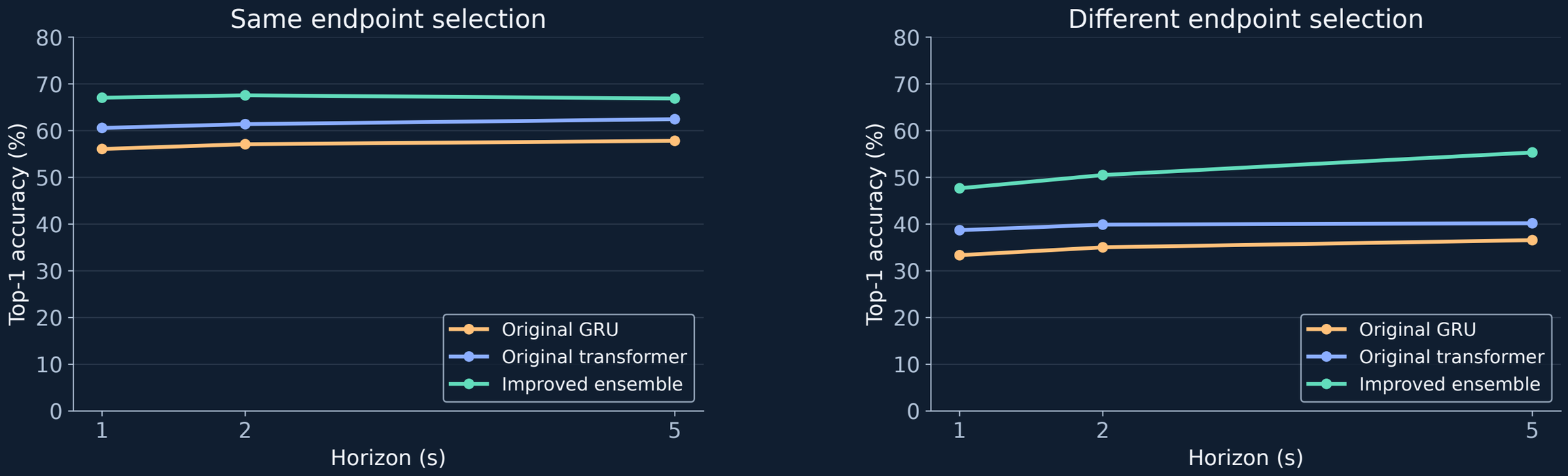
Horizon	Origins	Both correct	Error corrected	New error	Both wrong
+1 s	30,802	16,243	3,968	1,920	8,671
+2 s	30,466	15,872	3,983	1,914	8,697
+5 s	29,458	14,556	4,144	1,911	8,847

AT +5 SECONDS

4,144 corrections − 1,911 regressions = 2,233 additional correct forecasts.
8,847 origins are still wrong under both models.

Changed selections gain most; unchanged cases also matter

True current labels define these diagnostic subsets only; they are not inputs.

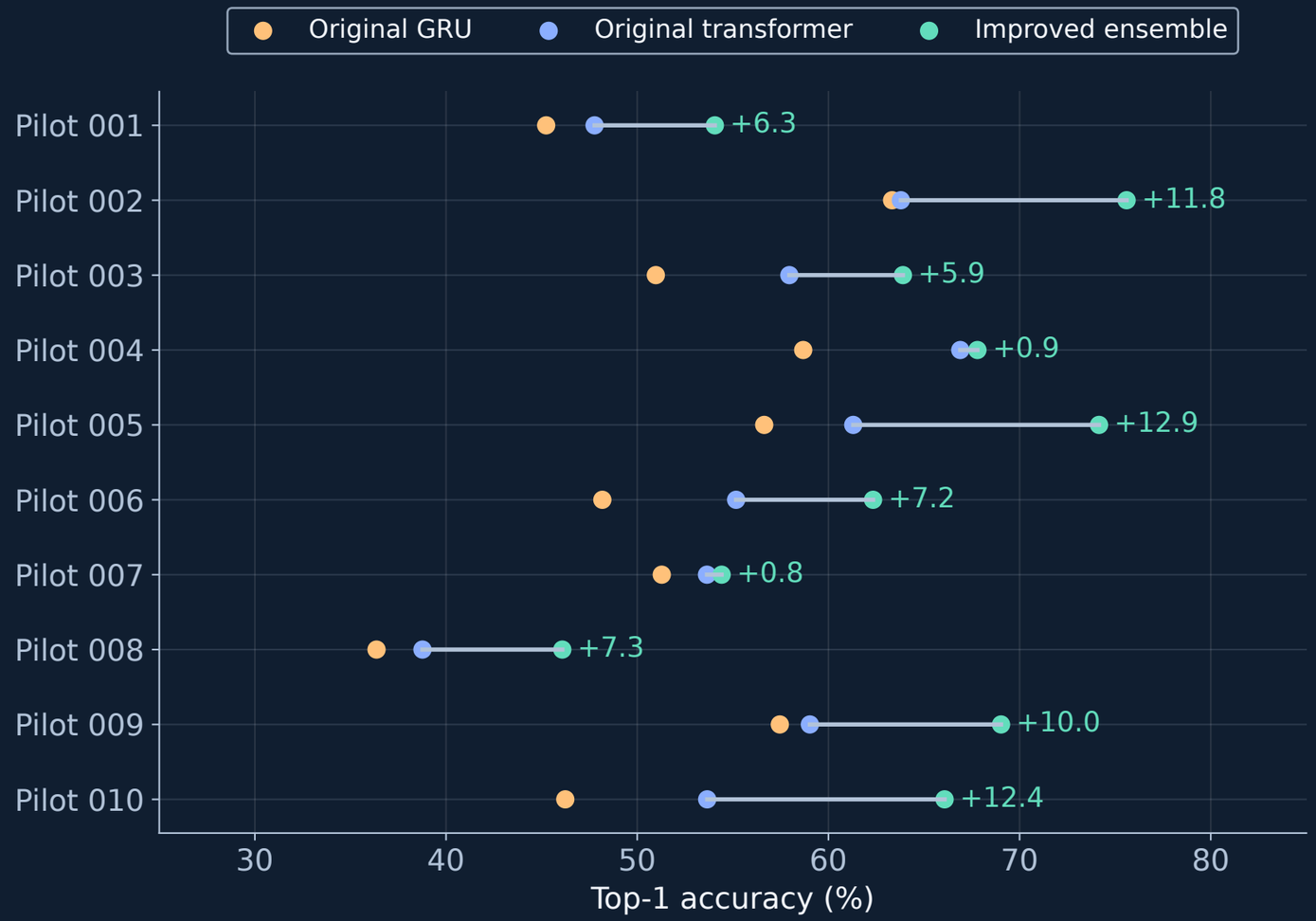


+5 s unchanged: 62.44% → 66.87% | changed: 40.18% → 55.33%

Unchanged +5 s top-3 falls 91.12% → 90.81%; its NLL worsens 1.0575 → 1.0813.

Every pilot improves; absolute difficulty still varies

+5-second top-1 accuracy for the known pilot population.



GAIN AT EVERY HORIZON

30 of 30 pilot / horizon cells improve.
At +5 s, gains range from
+0.77 to +12.86 points.

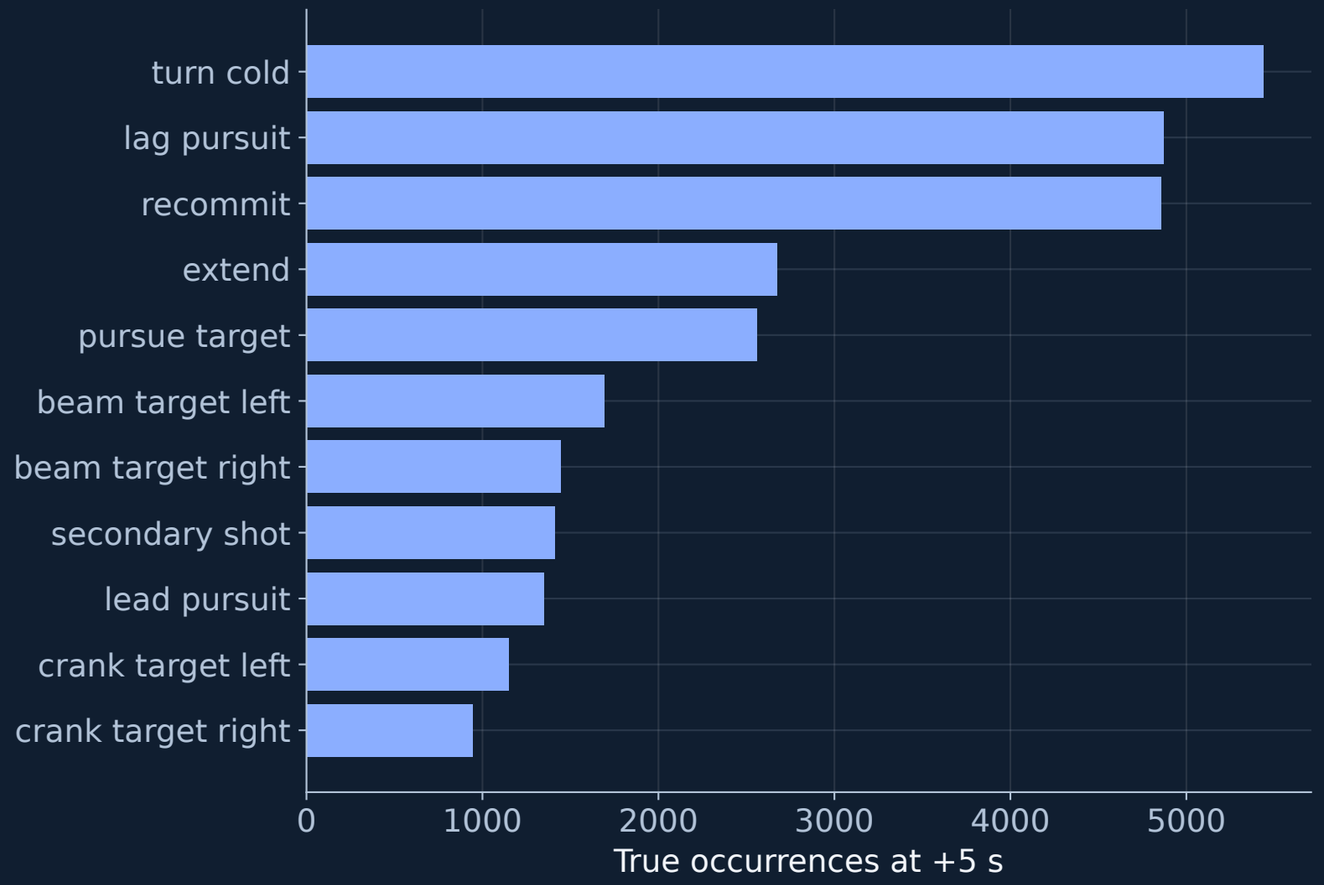
WHAT THIS DOES NOT MEASURE

Pilot competence or tactical quality.
Generalisation to unseen policies.
Significance for each pilot separately.

summary_metrics.csv | Per-pilot all cohort | Development validation; test reserved

A small set of labels dominates the validation windows

Class support explains why overall accuracy and macro-F1 tell different stories.



11 FREQUENT LABELS

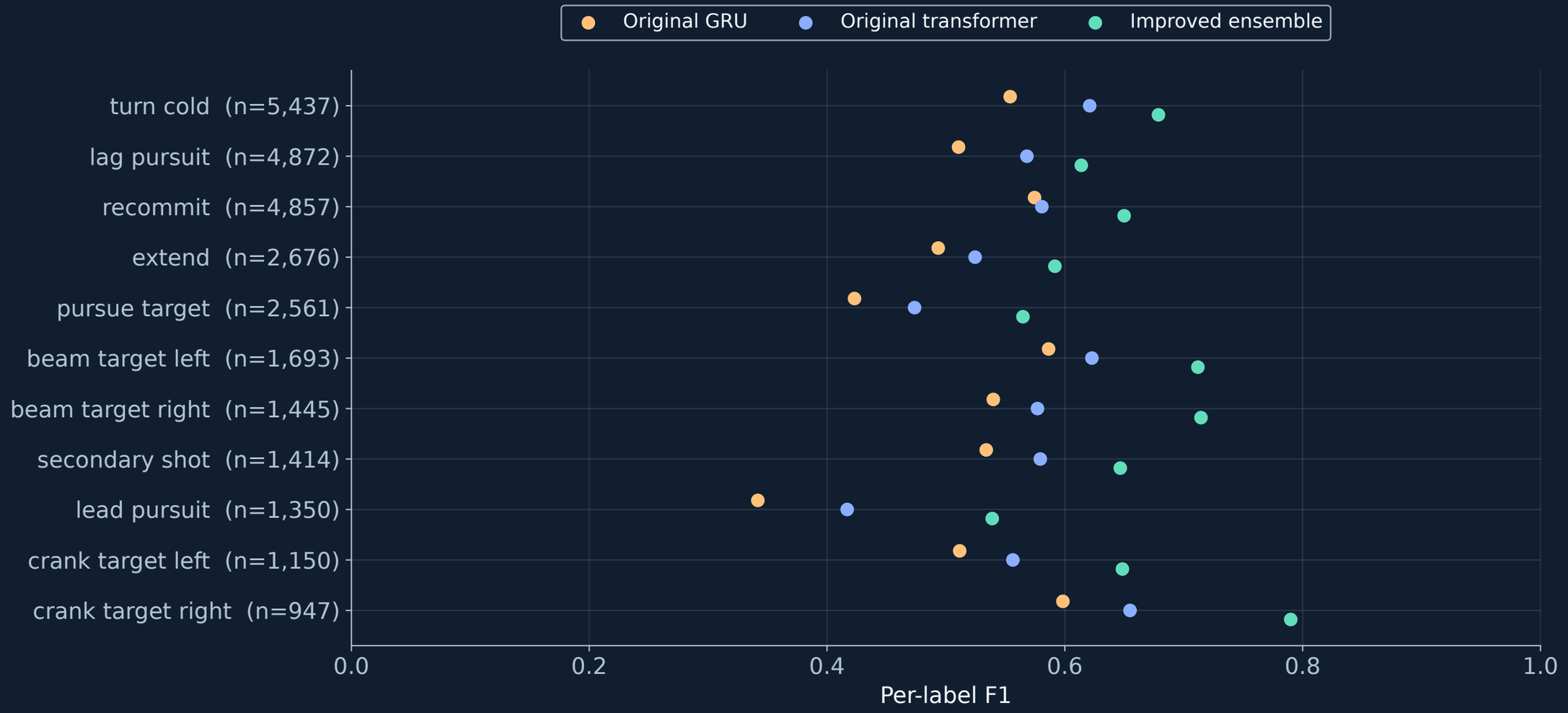
96.42% of valid windows.
At least 500 occurrences each.

THE REMAINDER

15 supported labels below 500.
7 entries with no truth support.
Rare-label estimates are fragile.

Frequent labels: compare F1 with the original models

All labels with at least 500 true +5 s occurrences; selected by support, not improvement.



Sparse labels: show the gains alongside the small counts

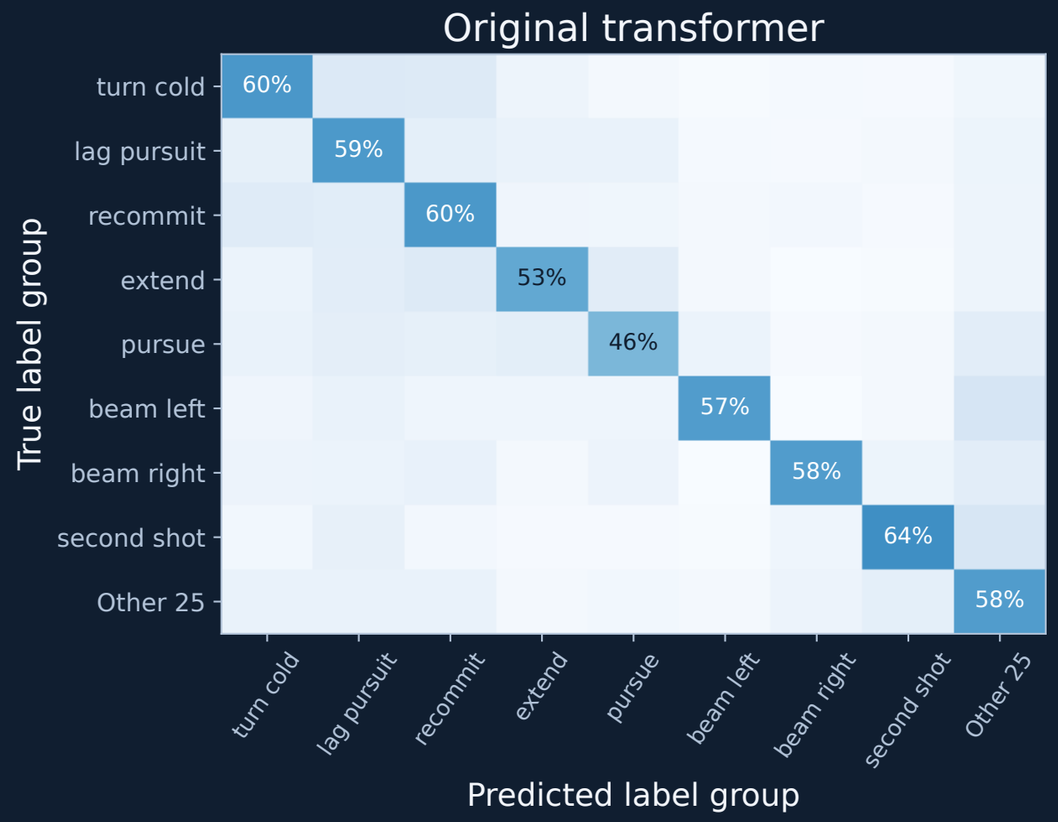
Every supported +5 s label below 500 occurrences. Do not infer reliable rare-class generalisation.

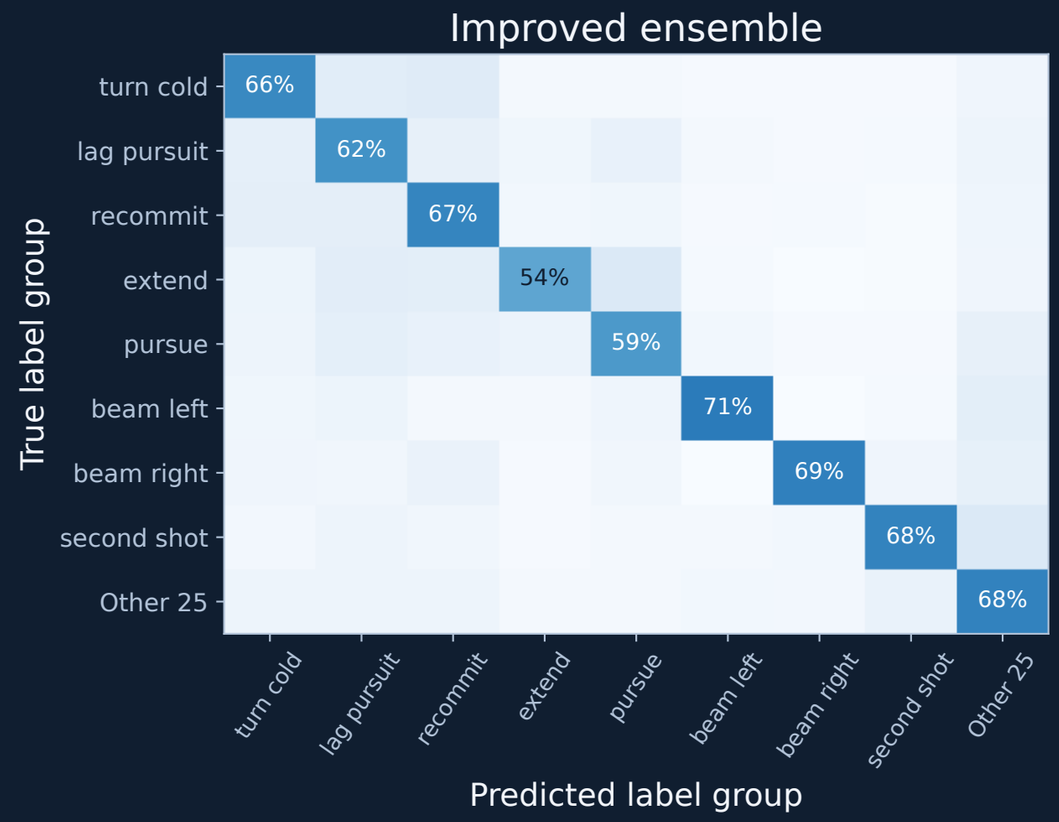
Label	Support	Old F1	New F1	New precision	New recall
short range attack	216	0.185	0.305	0.402	0.245
descend to altitude	162	0.480	0.619	0.528	0.747
launch	130	0.300	0.494	0.552	0.446
decelerate to speed	126	0.246	0.326	0.517	0.238
notch left	124	0.302	0.400	0.488	0.339
abort support	113	0.293	0.521	0.561	0.487
notch right	80	0.197	0.444	0.703	0.325
maintain heading	36	0.447	0.553	0.525	0.583
turn to heading	20	0.412	0.510	0.419	0.650
beam missile right	20	0.000	0.203	0.122	0.600
support missile	19	0.000	0.320	0.667	0.211
beam missile left	6	0.000	0.250	0.500	0.167
search	2	0.000	0.000	0.000	0.000
climb to altitude	1	0.000	0.000	0.000	0.000
defensive reversal	1	0.000	0.000	0.000	0.000

Three labels with only 1–2 true occurrences still have zero F1; seven further entries have zero support.

Confusion matrices reveal the remaining label mixtures

+5 s, row-normalized over all predictions. The eight most frequent true labels are shown individually.

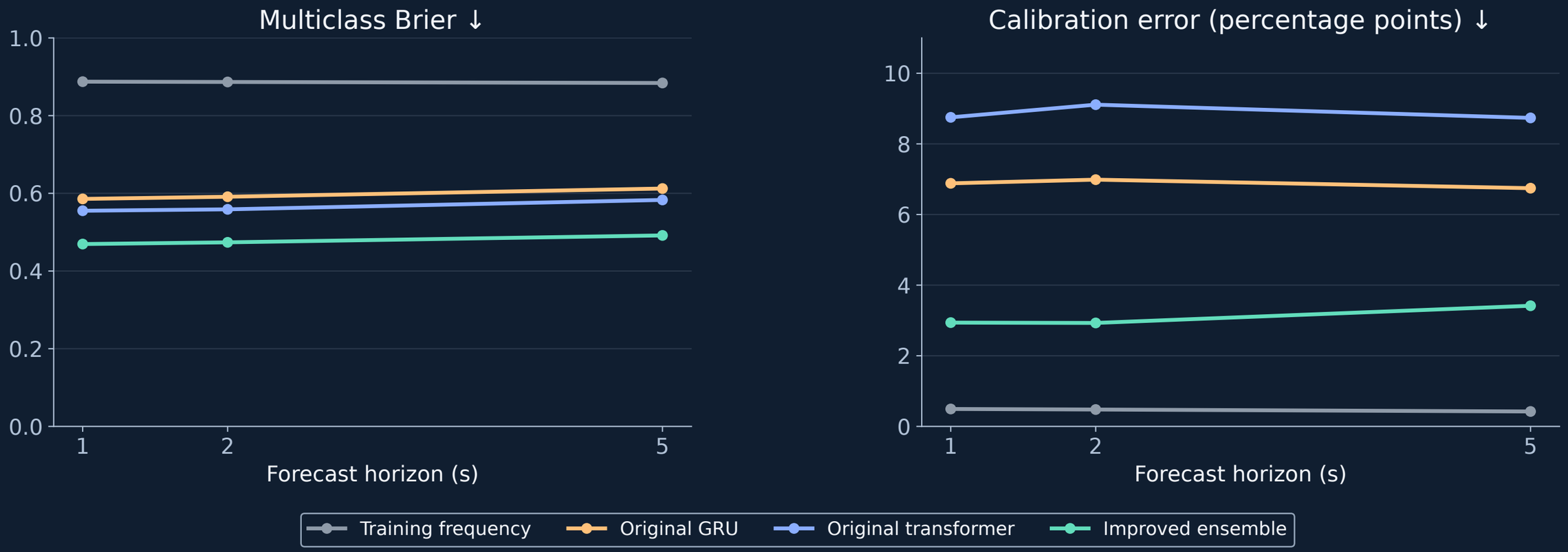




Darker = larger share of a true-label row. “Other 25” is for display only; scores retain all 33 labels.

The ensemble improves probability scores over both networks

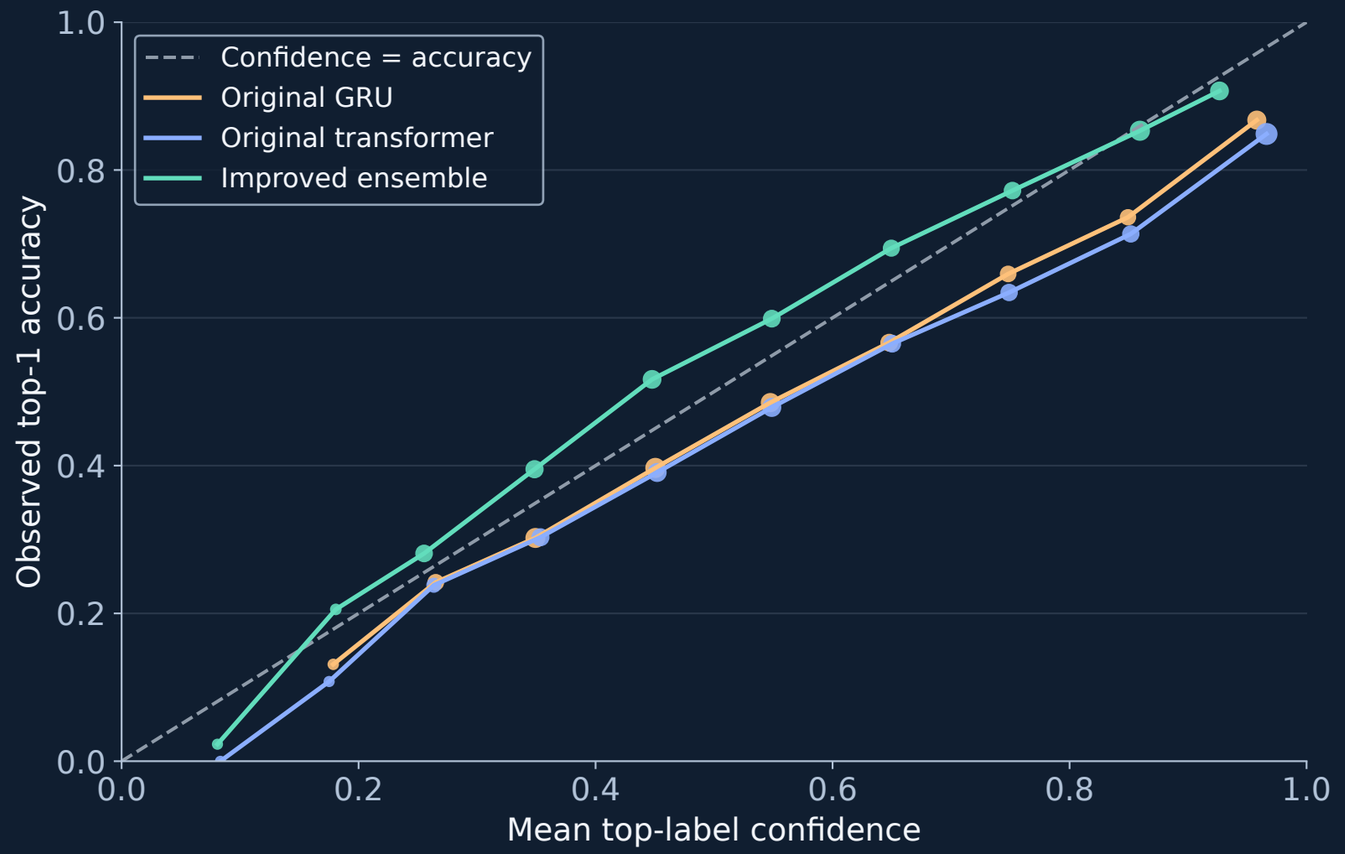
Brier measures the entire distribution; ECE measures binned top-label calibration.



Low ECE alone is not enough: the frequency reference has low ECE but poor accuracy.

Confidence is closer to observed accuracy after improvement

+5 s reliability diagram using ten fixed confidence bins; marker size reflects bin support.



AT +5 SECONDS

Original transformer
Mean confidence 64.64% / accuracy 55.90%

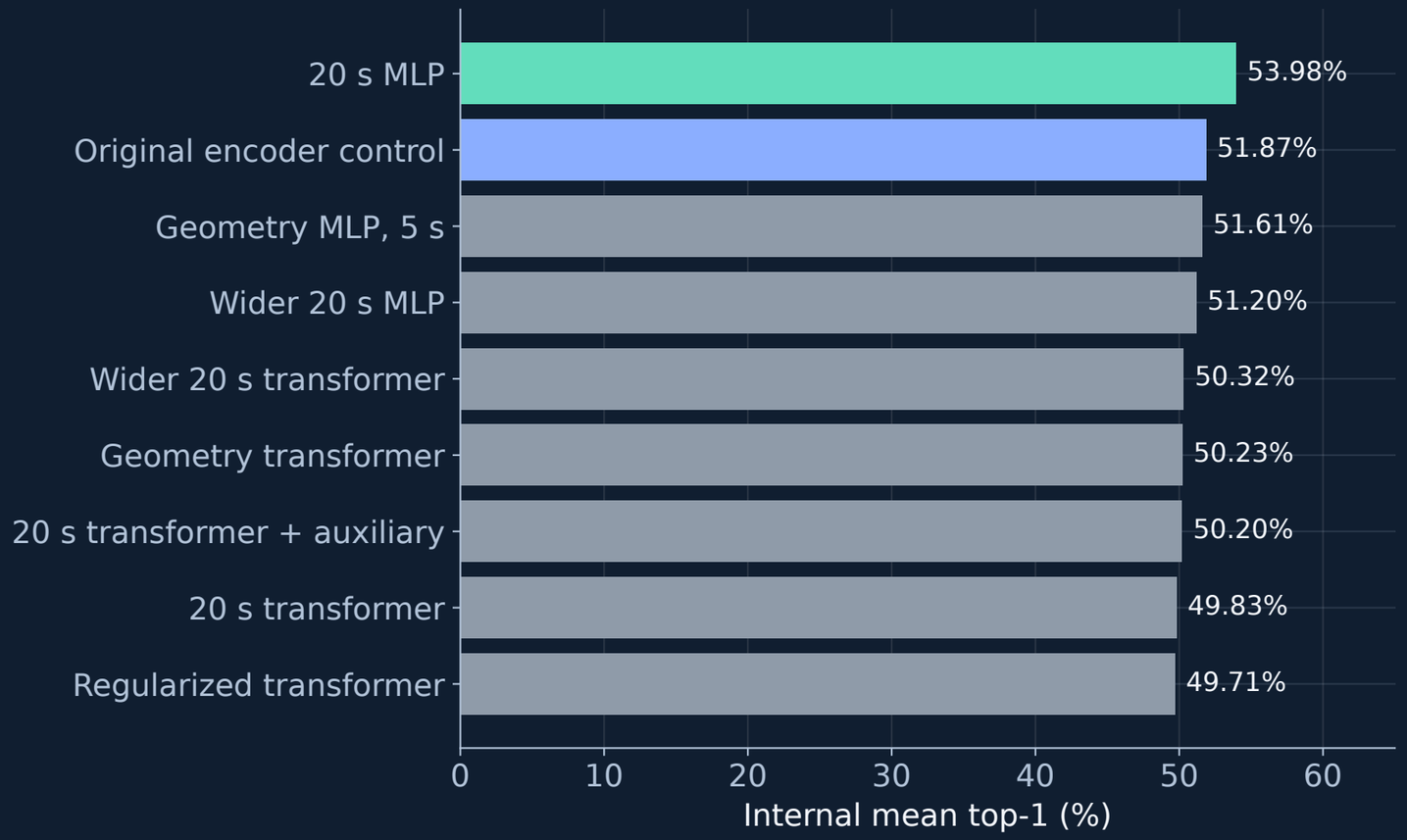
Improved ensemble
Mean confidence 60.83% / accuracy 63.48%

INTERPRETATION

Less overconfidence in aggregate.
Calibration still varies by confidence bin.

The development search also records unsuccessful changes

First internal screen: mean top-1 across horizons. These are not the outer-validation scores.



WHAT THE SCREEN SAYS

More geometry or a wider transformer did not reliably help. The 20 s MLP warranted follow-up.

WHAT IT DOES NOT SAY

An isolated causal contribution for every change. Settings and training procedures differ.

initial_search.csv | Internal split 0 | Development validation; test reserved

The final gain comes with more history and computation

Freeze the selected procedure before evaluating outer validation.

Step	Protocol
Fit / stop / select groups	147 / 21 / 42, inside the original 210 training groups
Confirmation	Two internal partitions; NLL and macro-F1 eligibility guards
Final refit	Four variants × seeds 42, 43, 44; fixed epochs and temperatures
Ensemble weights	Uniform arithmetic probability average across 12 members

ORIGINAL TRANSFORMER → ENSEMBLE

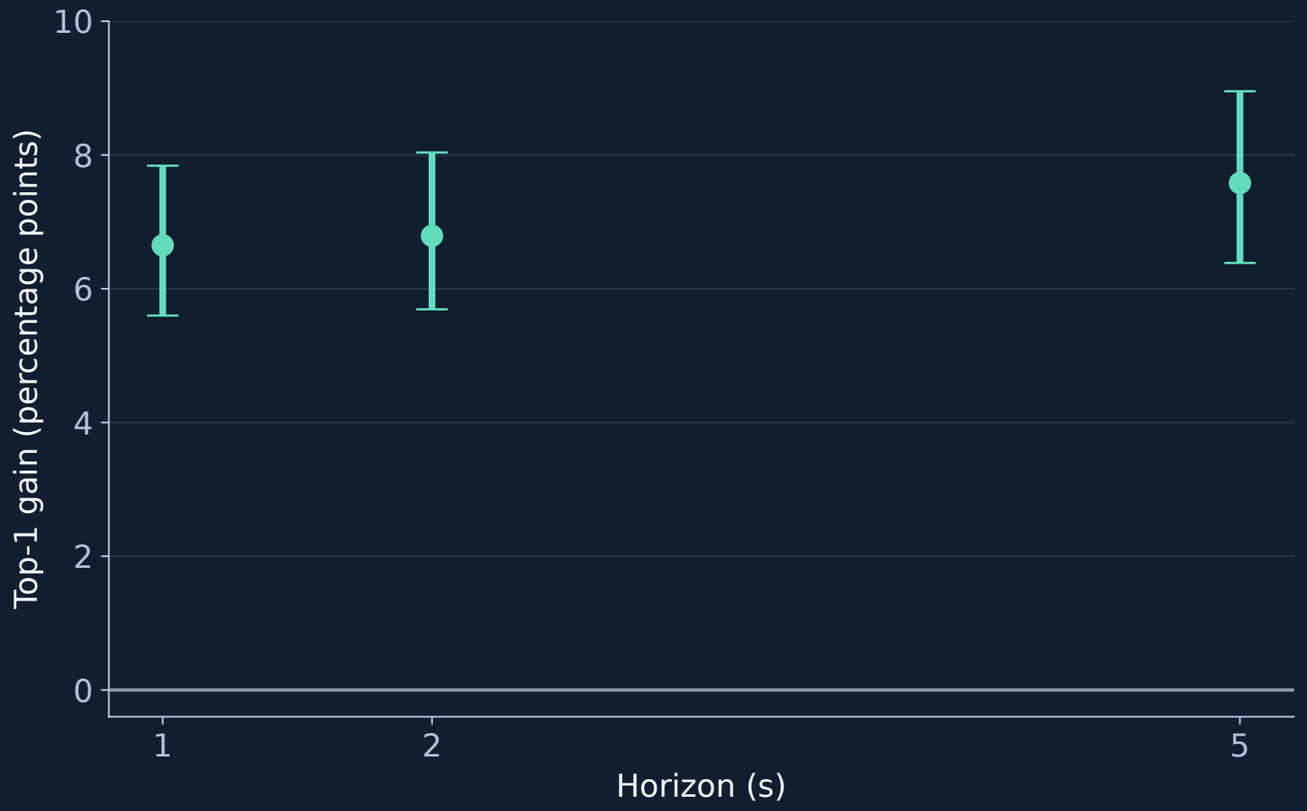
5 s → up to 60 s history
0.44M → 28.98M total parameters

A SMALLER ALTERNATIVE IS STILL PLAUSIBLE

Internal accuracy: 60.96% vs 60.98%.
Six versus twelve models was not decisive.

The gain remains positive within scenario-level intervals

Resample whole scenarios, preserving all ten pilots and overlapping windows.



PAIRED 95% INTERVALS

+1 s: +6.65 points [5.60, 7.84]
+2 s: +6.79 points [5.69, 8.04]
+5 s: +7.58 points [6.39, 8.95]

THE EVIDENCE LIMIT

45 scenario groups; 5,000 draws.
Original validation already informed baselines.
Reserved test remains unevaluated.

A stronger forecaster, with specific remaining limitations

Use the complete comparison to judge the gain—not only the headline accuracy.

WHAT IMPROVED

Top-1 and top-3 at every horizon.
Catalogue macro-F1 and probability scores.
All ten pilots; changed-selection accuracy.
Net corrections exceed new errors.

WHAT REMAINS UNRESOLVED

Rare and absent labels.
Some unchanged-subset metrics worsen.
Compute cost and minimal ensemble size.
Unseen pilots and realistic sensor inputs.

Next: freeze the comparison, evaluate the reserved test once, and report all metrics.

Complete scorecards, every catalogue entry and reproducible evidence accompany this deck.